

Notes on Unit –I

Linear Project Management Framework

Basics of Project Management

Introduction

- A farmer taking up crop cultivation
- A construction company constructing a bridge
- Indian Railways changing the meter gauge railway track to broad gauge
- An FMCG company introducing its products into a new virgin market
- A company hiring fresh graduates
- A student pursuing MBA

What is common to all these? All these are **projects**.

A project is not merely establishing an industry or constructing a building. It is just something new, something unique, planned and executed for good.

Definition of Project

- A project is a temporary endeavor undertaken to produce a unique product or service
 - Temporary – Definitive beginning and end
 - Unique – New undertaking, unfamiliar ground

PMBOK (Project Management Body of Knowledge) defines project as a **temporary endeavour undertaken to create a unique product or service**. **Temporary** means that every project has a definite end, and **Unique** means that the product or service is different from all similar products or services.

Turner defines projects as an endeavor in which human (or machine), materials, financial and knowledge resources are organized in a novel way, to undertake a unique scope of work of given specification, within constraints of cost and time, so as to deliver quantitative, qualitative, and consumer oriented product and service.

Bridgefield group defines project as a related set of activities and milestones with a preset goal and time frame that is designed as a specific event and not an ongoing process.

Project can also be defined as a single use plan to achieve a certain objective of introducing something unique or a change and ensure that progress is maintained in

line with the objective, generally in terms of time, cost, and various technical and quality performance parameters.

The following are the important aspects of a project:

- Starting date
- Specific goals and conditions
- Defined responsibilities
- Budget
- Planning
- Fixed end date
- Parties involved

Project Management

Project management is a methodical approach to planning and guiding project processes from start to finish. It is the method of planning the plan. It starts from project definitions and ends with goal achievement.

PMBOK defines project management as the application of knowledge, skill, tool and techniques to project activities in order to meet stakeholder's needs and expectations from a project.

Bridge group defines it as the methods and disciplines used to define goals, plan and monitor tasks and resources, identify and resolve issues, and control costs and budgets for a specific project.

Project Characteristics

The various characteristics of the project are

- **Fixed set of objectives:** The project starts when the objective(s) is finalized. The project comes to an end as soon as the objectives are attained.
- **Tenure:** Project is never a continuous activity, it has to come to an end. Its life span is fixed.
- **Team work:** It needs a team to accomplish various activities.
- **Unique:** All projects are unique in themselves, no two projects are exactly similar.
- **Life cycle:** Like all living organisms, project starts slowly (definition phase), then starts building up in size (planning phase), then reaches peak (implementation phase) before finally getting terminated.
- **Made to order:** The customer always decides the objective and informs the constraints like time and cost objectives

- **Single entity:** Generally, projects are the responsibilities of a single person/entity but certainly there are many participants in a project, who are helping the single entity in the accomplishment of project
- **Multi-skilled staff:** The staff needed for a project, including the project manager needs to have a wide range of skills including technical skills, human skills, financial skills, negotiation skills, etc.
- **Subcontracting:** Subcontracting is practically unavoidable in project management. As specialized knowledge or workforce is needed for a very small duration in a project, it is difficult and costly to employ or retain. Therefore, they are just hired for small duration or specific job from outside agency.
- **Risk and uncertainty:** Projects are risky as the activities involved in projects are non-retrievable. Thus, risk is unavoidable. However, risk can be reduced considerably using various forecasting techniques and project management and control tools

Importance of Project Management

What has led to increased usage of the concept of project management in recent times?

- **Rapidly changing technologies:** Technologies are changing very fast, so all manufacturing as well as service organizations have to cope up with technological changes, which provide a big scope for project management.
- **High entropy of the system:** Changes are very fast. So, energy levels are high. To adapt to the fast changing world, no organization can stick to old things or systems. Any modification or modernization leads to the need of project.
- **Squeezed life cycle of products:** Product life cycle is squeezed to a great extent with innovations taking place at a very rapid rate. Projects are needed for the upgradation of products.
- **Globalization impact:** All producers and service providers in the present world are exposed globally. They need to modify their system of operations to match the global practices, thus creating opportunity for projects.
- **Large organizations:** They face problems of management of huge workforce and work division, so they divide their work in projects and create a team to accomplish the objectives in the form of projects. This has also helped the organization to develop a method for performance appraisal.
- **Customer focus:** Increased customer focus has been a market trend in recent times. A few years back, cost reduction was a major formula of success for an

enterprise. Thus, there was more emphasis on standardization. In recent years, customer focus has redirected market towards customization. Though it is not purely customization, it is more of a combination of standardization and customization. All this has led to the application of project management

Objectives of Project management

There are four major objectives of project management

- **Scope:** Scope means what are the expectations from you as a project manager and your team. A civil contractor always has well-defined scope, like all civil works including excavation, foundation, concreting, brickwork, plastering of all walls as per the attached drawings.
- **Performance:** A project is always expected to have a well defined performance level. If a project is unable to adhere to the desired performance of a customer, it is certainly an unsuccessful project.
- **Time:** A successful project is the one which is completed within the time limits perceived during the planning. As the cost is dependent on time, time management becomes a crucial activity of project management.
- **Cost:** It is dependent on all the above objectives. Mathematically it can be written as: $\text{Cost} = f(P, T, S)$.

Therefore, cost is a function of performance, time and scope. If any of the above increases, it is surely going to increase the cost of the project.

Another approach in defining the objectives is the SMART approach.

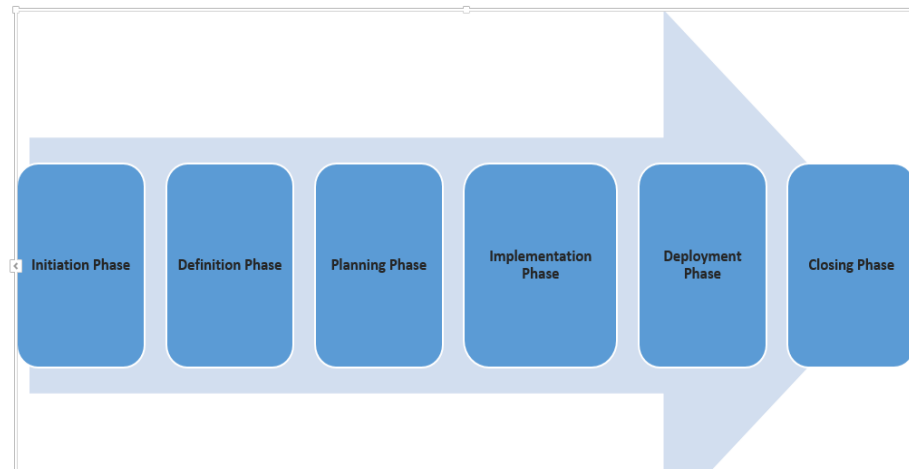
- **Specific:** Project should target a specific goal
- **Measurable:** It should be quantifiable
- **Attainable:** It should be attainable with resources available
- **Realistic:** It should be realistic in nature
- **Time Limit:** There should be fixed time limits

Key Areas of Project Management

- Scope Management
- Issue Management
- Cost Management
- Quality Management
- Communications Management
- Risk Management
- Change Control Management

Project Life Cycle

Project Life Cycle



Project life cycle and its phases

Project life cycle divides the sequence of operations of project in to different phases. Regardless of scope or complexity, any project goes through a series of stages during its life. Project activities must be grouped into phases to facilitate project manager and his team to plan and organize various inputs effectively. It also helps in identifying deviations and thus helps in decision making with regard to continuation or termination of the project.

Generally, there are four stages of project life cycle which are

2.1.1. Idea Generation (Concept Phase)

Anyone who is planning to invest starts searching everywhere for new ideas. One can start a new project by defining its objectives, scope, purpose and deliverables to be produced. He will also hire his project team, set up the project office and review the project, to gain approval to begin the next phase. The basic processes of this initiation phase are

- **Project document:** This is a statement describing the characteristics of the
- **Project feasibility document:** This contains constraints and alternative solutions. The four steps in the project feasibility study are:
 - Problem description
 - Approach to be used
 - Alternate generations for solving the problem
 - Preliminary recommendations
- **Project concept document:** It will answer the following questions
 - What is to be done?
 - How will it be done?
 - Why is it to be done?
- **Project charter:** Project charter formally communicates the initiation of the project. It consists of project scope, project authority and KSF (Key Success Factors)

During this phase, project team is responsible for the following activities:

- Conducting interviews with customers and stakeholders
- Conducting research for generating more necessary information.
- Preparing project feasibility document, project concept statement and project charter.

Project Planning Phase

Project planning phase follows the project initiation phase. Countless hours during the succeeding phases can be saved with proper planning.

The purpose of the project planning phase is to:

- Determine project requirements
- Decide project cost and schedules
- Search for sources of all resources

The basic processes of the project planning phase are:

- **Defining the scope:** Define the scope of the project and its limitations.
- **Preparing the work breakdown structure:** Divide the whole project into smaller activities
- **Role assignment:** Assign jobs to individuals or group of individuals as predefined activities or tasks.
- **Project scheduling:** Determine optimum schedule of the project and show it on a Gantt Chart.
- **Fund allocation:** Allocation of funds for individual activities

Other subsidiary processes in the planning stage are:

- **Risk management planning:** It includes identification of possible causes and effect of the risks and trying to reduce the impact of risk.
- **Procurement planning:** Decisions regarding all products, services or resources needed to accomplish the project.

In the planning stage, various steps are taken which includes:

- **Final techno-economic feasibility of the project:** This is the last chance for changing the decision, as after this stage, it proves too closely to shut down the project or change the project.
- **Basic engineering and process design:** The process is selected and basic engineering is done. The documents with respect to equipment specification are prepared.

- **Division of work/responsibilities:** Different activities are allocated to individuals or groups.
- **Identify potential vendors and subcontractors:** No project is complete without the help of outside expert agencies called subcontractors. The potential suppliers of various equipment, civil construction agencies and similar agencies are identified and negotiated.
- **Detailed engineering design:** based on the designs of equipment supplier, detailed engineering is performed. The final layout is prepared and the work schedule prepared.
- **Final estimation of the cost of the project:** The above steps leads to finalizing quite accurate cost of the project. This is essential as the next step would involve arrangement of funds.
- **Decision of capital structure and means of finance:** The final decision with respect to financing the project is needed during the planning phase. It is a crucial decision generally taken by core strategic group with the advice of finance managers.
- **Final schedule of implementation (next phase):** The next phase will be implementation. A proper schedule of implementation is essential to avoid confusions. The schedule of implementation tells all the members of the team when a particular activity should start and end. It will provide the milestones of every activity. The techniques used are PERT, CPM, Gantt chart, crashing resource allocation and resource leveling.

2.1.2. Implementation or Execution Phase

Project execution is characterized by the actual work on the tasks planned and project control involves the comparison of the actual performance with the planned performance and taking appropriate corrective action to get the desired output.

During this phase, project team is responsible for the following activities:

- The team members perform the tasks allocated in the earlier phase under the supervision of the project manager and report to him.
- Project manager is responsible for performance measurement, which includes finding variances with respect to cost, schedule and scope.
- Project manager is responsible for providing project status report to all key stakeholders. He should specifically inform the deviation from the plan to the stakeholders. He should also determine the root cause for the deviations and

suggest the alternate actions to encounter the deviation caused or expected. This helps stakeholders to decide the corrective action to be taken.

- All project key stakeholders are responsible for the review of the variances.
- All project key stakeholders are responsible for taking necessary action of the variances thus determined so as to complete the project within time and cost.

The basic process of the project execution can be:

- Execution of the project plan
- Handle the changes
- Project control

The subsidiary processes during project execution can be:

- Quality control
- Performance monitoring
- Project administration
- Risk monitoring and control
- Scope and control
- Schedule and cost control
- Management of outside agencies (subcontractors)

The key activities during this phase of execution include:

- **Award contracts to contractors, vendors, subcontractors:** Final selection of suppliers of various supplies of services (generally termed contractors) and physical equipment (generally termed vendors).
- **Procure equipment and services:** After continuously monitoring the suppliers, the project team has to procure the goods and services.
- **Erection of equipment:** The procured equipment needs to be placed on the designed place after preparing the required foundation.
- **Control and monitor project cost, schedule and scope:** As majority of efforts, time and cost are incurred during this phase, it is critical to monitor the project schedule and cost during this phase. This is generally done using various tools like Gantt chart and Earned Value Analysis.
- **Motivation of project team:** As this phase consumes maximum energy of the team members, motivating them during this phase is critical to the success of the project.

2.1.3. Termination Phase (Clean-up Phase)

The last step performed to say good bye to a project is the termination phase. The termination of a project is inevitable, but how it is terminated and when may have a profound and long lasting impact on the organization and its employees.

In the end, all projects, both successful and unsuccessful, will have to be terminated. During the termination phase, the project's resources are redistributed, financial records are closed, and project personnel are reassigned. The organization's sensitivity to the concerns of the project team can have a lasting impact on their commitment and productivity. Lastly, a final report, which discusses the project's successes and shortcomings, is prepared for senior management. This report can significantly influence how the organization manages projects in the future.

According to Meredith and Mantel (1995), there are three ways to terminate a project: extinction, inclusion, or integration. Termination by **extinction** means the project is completed. For example, the new project has been developed and given to the client, the building has been completed and accepted by the purchaser, or the software has been installed and is running.

By contrast, termination by **inclusion** is a very different process. The complete project team and its equipment are transferred to a new division. As one might expect, this type of change places significant additional stress on the day-to-day operations of the organization. Project managers and team members must be sensitive to these stresses until the organization is able to settle into a new and more stable routine.

The most common, but also the most complex, method of termination is by **integration**. The project's resources, personnel, and functions are absorbed as a part of the original organization. The major problem associated with this termination process is the ability of the organization to blend technological differences between the project and the organization. Past experience appears to play a key role in successfully integrating terminated projects.

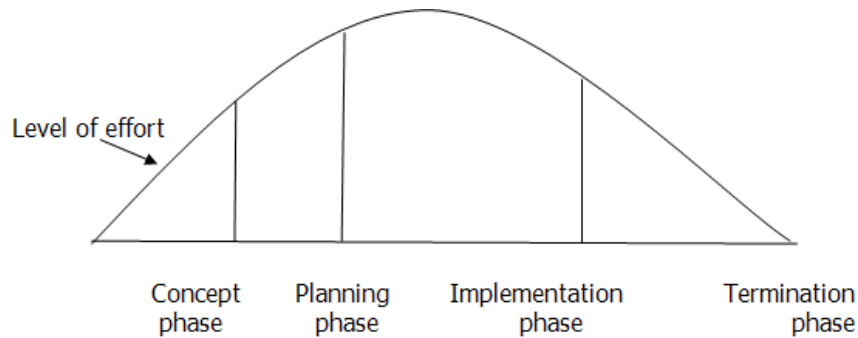


Figure 2.1 Level of effort in different phases of project

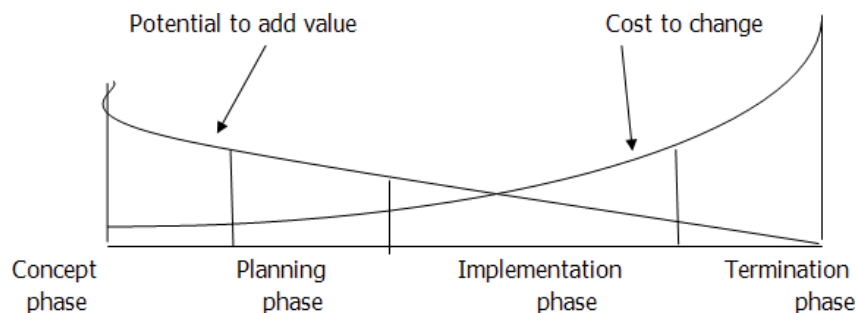


Figure 2.2: Cost of change and potential to add value during project life cycle

Project Audit

Ideally, a project audit should be conducted by an independent examiner, who can remain objective in the assessment of information.

Key activities during this phase are:

- **Ensure completion and acceptance:** The project manager should ensure that all project related activities are completed and the project is acceptable to all the stakeholders, including owners, customers and the new team for operations.
- **Prepare a final report:** A final report regarding the deviations from scope, cost and schedule should be submitted along with the operations manual and risk perceived during operations.
- **Ensure payments:** The project manager should ensure that all the payments to vendors, contractors and subcontractors have been done.
- **Assign personnel:** Some team members of the project should be assigned duties in operations management team, which will help the new team in managing operations smoothly, as it is not a totally new team for the management then.

A Risk Management Plan :A defined and documented process agreed upon by project stakeholders for how risks will be identified, assessed, a decision made on mitigation (or if the risks will be accepted), how a response plan will be developed and what controls will be put in place to monitor risks over the duration of the project.

Identify Risks – Capture Form

A way to efficiently capture identified project risks and add to the Register.

Risk Register

A log of identified risks and their status.

Qualitative / Quantitative Analysis Tools

Methods for analyzing / evaluating the probability and impact of risks on the project objectives.

Response / Mitigation Plan

Determine if the risks are acceptable or not based on assessment and plan for mitigation.

Control Risks

Assess effectiveness through methods like risk audits and continually improve your project execution.

Risk Register Example

A typical register may look like the table below. Risks are added to the register as they are identified and the impact and probability of occurrence are assessed through qualitative and quantitative methods.

Risk Identified	Probability	Impact	Action to Prevent / Manage Risk
Failure to retain key project staff	Low	High	Establish team with written commitment to timeline, ensure knowledge captured through project

Qualitative Risk Assessment

Qualitative assessment is a *subjective* evaluation of factors visualized in a heat map and prioritized based on probability and impact. Rating scales are pre-defined and impact scales are tailored to the organization and the project objectives.

Quantitative Risk Assessment

Qualitative assessment is an *objective* numerical probabilistic assessment of impact and probability of factors. There are many methodologies available – some examples include:

- Critical Path Method

- Fault Tree Analysis
- Monte Carlo Simulation
- Sensitivity Analysis
- FMEA

Probability and Impact Matrix (Heat Map)

A typical matrix may look like the table below – example probability and impact definitions are shown:

Probability	Threats					Opportunities				
	Risk Score = Probability x Impact					High (RED) / Med (YEL) / Low (GRN)				
0.90 Very Likely	0.05	0.09	0.18	0.38	0.72	High	High	High	Med	Low
0.70 Likely	0.04	0.07	0.14	0.28	0.56	High	High	Med	Med	Low
0.50 Possible	0.03	0.05	0.10	0.12	0.40	High	High	Med	Low	Low
0.30 Unlikely	0.02	0.03	0.06	0.12	0.24	High	Med	Med	Low	Low
0.10 Very Unlikely	0.01	0.01	0.02	0.04	0.08	Med	Low	Low	Low	Low
	0.05	0.10	0.20	0.40	0.80	Very High	High	Med.	Low	Very Low
Example Impact Definitions – May Be Tailored to Each Project Objective Impact on an Objective (e.g. Cost, Schedule, Scope, Quality)										

Risks on the register are scored to determine impact on the project and what (if anything) needs to be done to mitigate and control the risks. Risks with a mitigation and control plan are logged on a Response Matrix.

The colour coding on the Heat Map indicates the severity of the risks:

- HIGH = Severe and likely to happen
- MEDIUM = Moderate risk; impact is not so severe
- LOW = Low risk of occurrence, low project impact if it does occur

Risk Mitigation

The objective of risk mitigation is to reduce the probability and/or consequences of a risk event to an acceptable threshold and define appropriate response. Questions To Ask:

- What are the available options?
- Tradeoffs (cost / benefit) of each option?
- Impacts of current decisions on options?

Risk mitigation actions may be costly and time consuming; actions taken are balanced against priority level of the risk. Organizations typically transfer risk where possible, for example through

product warranty.

Low-risk factors may be recognized by the Organization but absorbed as a matter of policy.

Risk Response

A typical response matrix may look like the table below:

Risk Event	Response	Contingency Plan	Trigger	Responsible
Equipment Malfunction	Mitigate: vendor change Transfer: warranty	Replace	Failure	J. Smith

RMMM Plan :

A risk management technique is usually seen in the software Project plan. This can be divided into Risk Mitigation, Monitoring, and Management Plan (RMMM). In this plan, all works are done as part of risk analysis. As part of the overall project plan project manager generally uses this RMMM plan.

In some software teams, risk is documented with the help of a Risk Information Sheet (RIS). This RIS is controlled by using a database system for easier management of information i.e creation, priority ordering, searching, and other analysis. After documentation of RMMM and start of a project, risk mitigation and monitoring steps will start.

Risk Mitigation :

It is an activity used to avoid problems (Risk Avoidance).

Steps for mitigating the risks as follows.

1. Finding out the risk.
2. Removing causes that are the reason for risk creation.
3. Controlling the corresponding documents from time to time.
4. Conducting timely reviews to speed up the work.

Risk Monitoring :

It is an activity used for project tracking.

It has the following primary objectives as follows.

1. To check if predicted risks occur or not.
2. To ensure proper application of risk aversion steps defined for risk.
3. To collect data for future risk analysis.
4. To allocate what problems are caused by which risks throughout the project.

Risk Management and planning :

It assumes that the mitigation activity failed and the risk is a reality. This task is done by Project manager when risk becomes reality and causes severe problems. If the project manager effectively uses project mitigation to remove risks successfully then it is easier to manage the risks. This shows that the response that will be taken for each risk by a manager. The main objective of the

risk management plan is the risk register. This risk register describes and focuses on the predicted threats to a software project.

Example:

Let us understand RMMM with the help of an example of high staff turnover.

Risk Mitigation:

To mitigate this risk, project management must develop a strategy for reducing turnover. The possible steps to be taken are:

- Meet the current staff to determine causes for turnover (e.g., poor working conditions, low pay, competitive job market).
- Mitigate those causes that are under our control before the project starts.
- Once the project commences, assume turnover will occur and develop techniques to ensure continuity when people leave.
- Organize project teams so that information about each development activity is widely dispersed.
- Define documentation standards and establish mechanisms to ensure that documents are developed in a timely manner.
- Assign a backup staff member for every critical technologist.

Risk Monitoring:

As the project proceeds, risk monitoring activities commence. The project manager monitors factors that may provide an indication of whether the risk is becoming more or less likely. In the case of high staff turnover, the following factors can be monitored:

- General attitude of team members based on project pressures.
- Interpersonal relationships among team members.
- Potential problems with compensation and benefits.
- The availability of jobs within the company and outside it.

Risk Management:

Risk management and contingency planning assumes that mitigation efforts have failed and that the risk has become a reality. Continuing the example, the project is well underway, and a number of people announce that they will be leaving. If the mitigation strategy has been followed, backup is available, information is documented, and knowledge has been dispersed across the team. In addition, the project manager may temporarily refocus resources (and readjust the project schedule) to those functions that are fully staffed, enabling newcomers who must be added to the team to “get up to the speed”.

Drawbacks of RMMM:

- It incurs additional project costs.
- It takes additional time.
- For larger projects, implementing an RMMM may itself turn out to be another tedious project.
- RMMM does not guarantee a risk-free project, infact, risks may also come up after the project is delivered.

Project Evaluation and Review Technique (PERT) :

PERT is appropriate technique which is used for the projects where the time required or needed to complete different activities are not known. PERT is majorly applied for scheduling, organization and integration of different tasks within a project. It provides the blueprint of project and is efficient technique for project evaluation .

2. Critical Path Method (CPM) :

CPM is a technique which is used for the projects where the time needed for completion of project is already known. It is majorly used for determining the approximate time within which a project can be completed. Critical path is the largest path in project management which always provide minimum time taken for completion of project

Difference between PERT and CPM :

S.No.	PERT	CPM
1.	PERT is that technique of project management which is used to manage uncertain (i.e., time is not known) activities of any project.	CPM is that technique of project management which is used to manage only certain (i.e., time is known) activities of any project.
2.	It is event oriented technique which means that network is constructed on the basis of event.	It is activity oriented technique which means that network is constructed on the basis of activities.
		3. It is a probability model.
		It is a deterministic model.
4.	It majorly focuses on time as meeting time target or estimation of percent completion is more important.	It majorly focuses on Time-cost trade off as minimizing cost is more important.
5.	It is appropriate for	It is appropriate for

S.No.	PERT	CPM		
	high precision time estimation.	reasonable time estimation.		
6.	It has Non-repetitive nature of job.	It has repetitive nature of job.		
7.	There is no chance of crashing as there is no certainty of time.	There may be crashing because of certain time boundation.		
8.	It doesn't use any dummy activities.	It uses dummy activities for representing sequence of activities.	9.	It is suitable for projects which required research and development.
				It is suitable for construction projects.

Advantages of Critical Path Method (CPM):

It has the following advantages:

1. It figures out the activities which can run parallel to each other.
2. It helps the project manager in identifying the most critical elements of the project.
3. It gives a practical and disciplined base which helps in determining how to reach the objectives.
4. CPM is effective in new project management.
5. CPM can strengthen a team perception if it is applied properly.
6. CPM provides demonstration of dependencies which helps in the scheduling of individual activities.
7. It shows the activities and their outcomes as a network diagram.
8. It gives a fair and concise procedure of documenting of project.
9. It helps in determining the slack time.
10. An explicit and clear approach of communicating project plans, schedules, time and cost performance is developed.
11. It is extensively used in industry.
12. It helps in optimization by determining the project duration.

Disadvantages of Critical Path Method (CPM):

It has the following disadvantages:

1. The scheduling of personnel is not handled by the CPM.
2. In CPM, it is difficult to estimate the completion time of an activity.
3. The critical path is not always clear in CPM.

4. For bigger projects, CPM networks can be complicated too.
5. It also does not handle the scheduling of the resource allocation.
6. In CPM, critical path needs to be calculated precisely.

Worked Examples on CPM Example 1 Determine the early start and late start in respect of all node points and identify critical path for the following network.

Introduction to CPM / PERT Techniques

Applications of CPM / PERT

Basic Steps in PERT / CPM

Frame work of PERT/CPM

Network Diagram Representation

Rules for Drawing Network Diagrams

Common Errors in Drawing Networks

Advantages and Disadvantages

Critical Path in Network Analysis

Introduction to CPM / PERT Techniques

CPM/PERT or Network Analysis as the technique is sometimes called, developed along two parallel streams, one industrial and the other military.

CPM (Critical Path Method) was the discovery of M.R.Walker of E.I.Du Pont de Nemours & Co. and J.E.Kelly of Remington Rand, circa 1957. The computation was designed for the UNIVAC-I computer. The first test was made in 1958, when CPM was applied to the construction of a new chemical plant. In March 1959, the method was applied to maintenance shut-down at the Du Pont works in Louisville, Kentucky. Unproductive time was reduced from 125 to 93 hours.

PERT (Project Evaluation and Review Technique) was devised in 1958 for the POLARIS missile program by the Program Evaluation Branch of the Special Projects office of the U.S.Navy, helped by the Lockheed Missile Systems division and the

Consultant firm of Booz-Allen & Hamilton. The calculations were so arranged so that they could be carried out on the IBM Naval Ordinance Research Computer (NORC) at Dahlgren, Virginia.

The methods are essentially **network-oriented techniques** using the same principle. PERT and CPM are basically time-oriented methods in the sense that they both lead to determination of a time schedule for the project. The significant difference between two approaches is that the time estimates for the different activities in CPM were assumed to be **deterministic** while in PERT these are described **probabilistically**. These techniques are referred as **project scheduling** techniques.

In **CPM** activities are shown as a network of precedence relationships using activity-on-node network construction

- Single estimate of activity time
- Deterministic activity times

USED IN: Production management - for the jobs of repetitive in nature where the activity time estimates can be predicted with considerable certainty due to the existence of past experience.

In **PERT** activities are shown as a network of precedence relationships using activity-on-arrow network construction

- Multiple time estimates
- Probabilistic activity times

USED IN: Project management - for non-repetitive jobs (research and development work), where the time and cost estimates tend to be quite uncertain. This technique uses probabilistic time estimates.

Benefits of PERT/CPM

- Useful at many stages of project management
- Mathematically simple

- Give critical path and slack time
- Provide project documentation
- Useful in monitoring costs

Limitations of PERT/CPM

- Clearly defined, independent and stable activities
- Specified precedence relationships
- Over emphasis on critical paths

2.2 Applications of CPM / PERT

These methods have been applied to a wide variety of problems in industries and have found acceptance even in government organizations. These include

- Construction of a dam or a canal system in a region
- Construction of a building or highway
- Maintenance or overhaul of airplanes or oil refinery
- Space flight
- Cost control of a project using PERT / COST
- Designing a prototype of a machine
- Development of supersonic planes

2.3 Basic Steps in PERT / CPM

Project scheduling by PERT / CPM consists of four main steps

1. Planning

- The planning phase is started by splitting the total project in to small projects. These smaller projects in turn are divided into activities and are analyzed by the department or section.
- The relationship of each activity with respect to other activities are defined and established and the corresponding responsibilities and the authority are also stated.
- Thus the possibility of overlooking any task necessary for the completion of the project is reduced substantially.

2. Scheduling

- The ultimate objective of the scheduling phase is to prepare a time chart showing the start and finish times for each activity as well as its relationship to other activities of the project.
- Moreover the schedule must pinpoint the critical path activities which require special attention if the project is to be completed in time.
- For non-critical activities, the schedule must show the amount of slack or float times which can be used advantageously when such activities are delayed or when limited resources are to be utilized effectively.

3. Allocation of resources

- Allocation of resources is performed to achieve the desired objective. A resource is a physical variable such as labour, finance, equipment and space which will impose a limitation on time for the project.
- When resources are limited and conflicting, demands are made for the same type of resources a systematic method for allocation of resources become essential.
- Resource allocation usually incurs a compromise and the choice of this compromise depends on the judgment of managers.

4. Controlling

- The final phase in project management is controlling. Critical path methods facilitate the application of the principle of management by expectation to identify areas that are critical to the completion of the project.
- By having progress reports from time to time and updating the network continuously, a better financial as well as technical control over the project is exercised.
- Arrow diagrams and time charts are used for making periodic progress reports. If required, a new course of action is determined for the remaining portion of the project.

2.4 The Framework for PERT and CPM

Essentially, there are six steps which are common to both the techniques. The procedure is listed below:

- I. Define the Project and all of its significant activities or tasks. The Project (made up of several tasks) should have only a single start activity and a single finish activity.
- II. Develop the relationships among the activities. Decide which activities must precede and which must follow others.
- III. Draw the "Network" connecting all the activities. Each Activity should have unique event numbers. Dummy arrows are used where required to avoid giving the same numbering to two activities.
- IV. Assign time and/or cost estimates to each activity
- V. Compute the longest time path through the network. This is called the critical path.

VI. Use the Network to help plan, schedule, and monitor and control the project.

The Key Concept used by CPM/PERT is that a small set of activities, which make up the longest path through the activity network control the entire project. If these "critical" activities could be identified and assigned to responsible persons, management resources could be optimally used by concentrating on the few activities which determine the fate of the entire project.

Non-critical activities can be replanned, rescheduled and resources for them can be reallocated flexibly, without affecting the whole project.

Five useful questions to ask when preparing an activity network are:

- Is this a Start Activity?
- Is this a Finish Activity?
- What Activity Precedes this?
- What Activity Follows this?
- What Activity is Concurrent with this?

2.5 Network Diagram Representation

In a network representation of a project certain definitions are used

1. Activity

Any individual operation which utilizes resources and has an end and a beginning is called activity. An arrow is commonly used to represent an activity with its head indicating the direction of progress in the project. These are classified into four categories

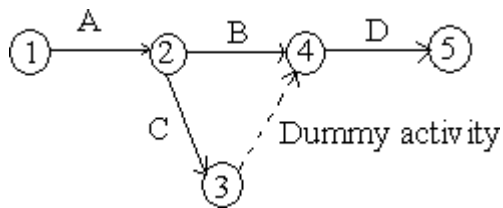
1. **Predecessor activity** – Activities that must be completed immediately prior to the start of another activity are called predecessor activities.

2. **Successor activity** – Activities that cannot be started until one or more of other activities are completed but immediately succeed them are called successor activities.
3. **Concurrent activity** – Activities which can be accomplished concurrently are known as concurrent activities. It may be noted that an activity can be a predecessor or a successor to an event or it may be concurrent with one or more of other activities.
4. **Dummy activity** – An activity which does not consume any kind of resource but merely depicts the technological dependence is called a dummy activity.

The dummy activity is inserted in the network to clarify the activity pattern in the following two situations

- To make activities with common starting and finishing points distinguishable
- To identify and maintain the proper precedence relationship between activities that is not connected by events.

For example, consider a situation where A and B are concurrent activities. C is dependent on A and D is dependent on A and B both. Such a situation can be handled by using a dummy activity as shown in the figure.



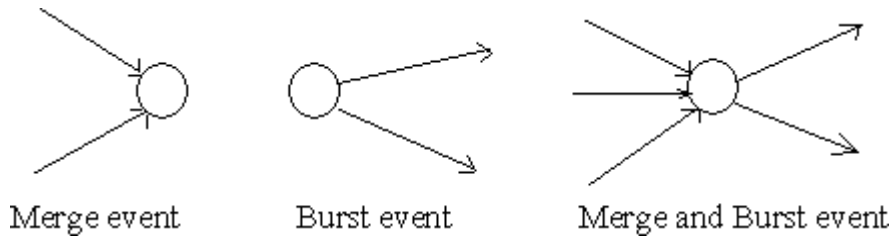
2. Event

An event represents a point in time signifying the completion of some activities and the beginning of new ones. This is usually represented by a circle in a network which is also called a node or connector.

The events are classified in to three categories

1. **Merge event** – When more than one activity comes and joins an event such an event is known as merge event.

2. **Burst event** – When more than one activity leaves an event such an event is known as burst event.
3. **Merge and Burst event** – An activity may be merge and burst event at the same time as with respect to some activities it can be a merge event and with respect to some other activities it may be a burst event.



3. Sequencing

The first prerequisite in the development of network is to maintain the precedence relationships. In order to make a network, the following points should be taken into considerations

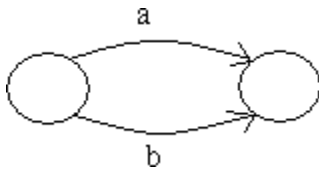
- What job or jobs precede it?
- What job or jobs could run concurrently?
- What job or jobs follow it?
- What controls the start and finish of a job?

Since all further calculations are based on the network, it is necessary that a network be drawn with full care.

2.6 Rules for Drawing Network Diagram

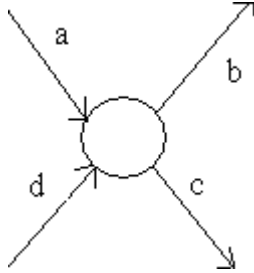
Rule 1

Each activity is represented by one and only one arrow in the network



Rule 2

No two activities can be identified by the same end events

**Rule 3**

In order to ensure the correct precedence relationship in the arrow diagram, following questions must be checked whenever any activity is added to the network

- What activity must be completed immediately before this activity can start?
- What activities must follow this activity?
- What activities must occur simultaneously with this activity?

In case of large network, it is essential that certain good habits be practiced to draw an easy to follow network

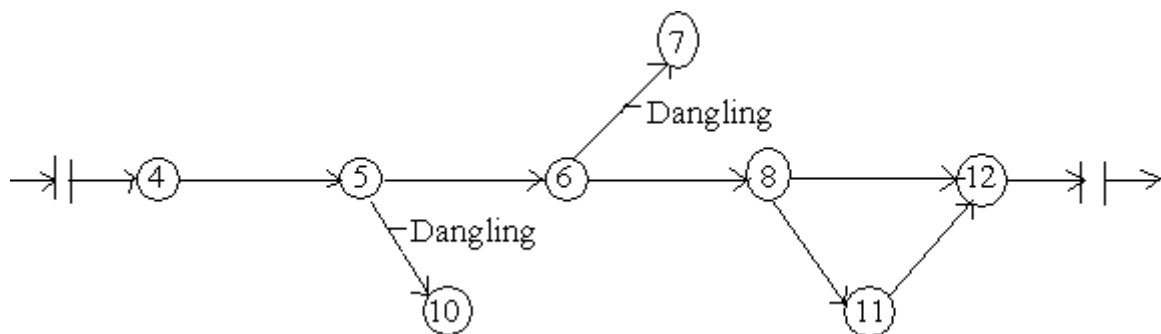
- Try to avoid arrows which cross each other
- Use straight arrows
- Do not attempt to represent duration of activity by its arrow length
- Use arrows from left to right. Avoid mixing two directions, vertical and standing arrows may be used if necessary.
- Use dummies freely in rough draft but final network should not have any redundant dummies.
- The network has only one entry point called start event and one point of emergence called the end event.

2.7 Common Errors in Drawing Networks

The three types of errors are most commonly observed in drawing network diagrams

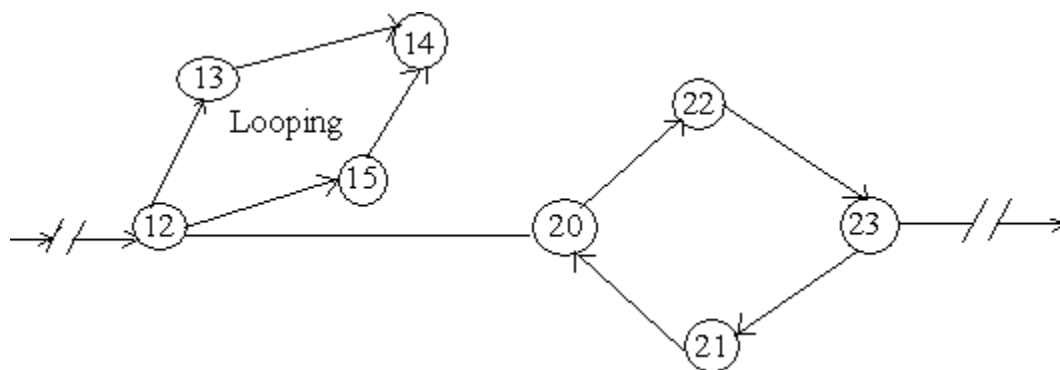
1. Dangling

To disconnect an activity before the completion of all activities in a network diagram is known as dangling. As shown in the figure activities (5 – 10) and (6 – 7) are not the last activities in the network. So the diagram is wrong and indicates the error of dangling



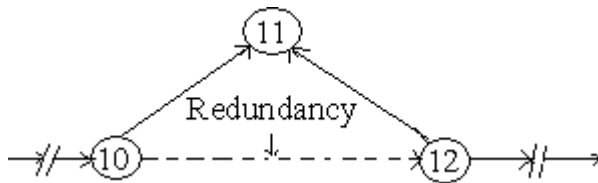
2. Looping or Cycling

Looping error is also known as cycling error in a network diagram. Drawing an endless loop in a network is known as error of looping as shown in the following figure.



3. Redundancy

Unnecessarily inserting the dummy activity in network logic is known as the error of redundancy as shown in the following diagram



2.8 Advantages and Disadvantages

PERT/CPM has the following advantages

- A PERT/CPM chart explicitly defines and makes visible dependencies (precedence relationships) between the elements,
- PERT/CPM facilitates identification of the critical path and makes this visible,
- PERT/CPM facilitates identification of early start, late start, and slack for each activity,
- PERT/CPM provides for potentially reduced project duration due to better understanding of dependencies leading to improved overlapping of activities and tasks where feasible.

PERT/CPM has the following disadvantages:

- There can be potentially hundreds or thousands of activities and individual dependency relationships,
- The network charts tend to be large and unwieldy requiring several pages to print and requiring special size paper,
- The lack of a timeframe on most PERT/CPM charts makes it harder to show status although colours can help (e.g., specific colour for completed nodes),
- When the PERT/CPM charts become unwieldy, they are no longer used to manage the project.

2.9 Critical Path in Network Analysis

Basic Scheduling Computations

The notations used are

(i, j) = Activity with tail event i and head event j

E_i = Earliest occurrence time of event i

L_j = Latest allowable occurrence time of event j

D_{ij} = Estimated completion time of activity (i, j)

$(Es)_{ij}$ = Earliest starting time of activity (i, j)

$(Ef)_{ij}$ = Earliest finishing time of activity (i, j)

$(Ls)_{ij}$ = Latest starting time of activity (i, j)

$(Lf)_{ij}$ = Latest finishing time of activity (i, j)

The procedure is as follows

1. Determination of Earliest time (E_j): Forward Pass computation

- **Step 1**

The computation begins from the start node and move towards the end node. For easiness, the forward pass computation starts by assuming the earliest occurrence time of zero for the initial project event.

- **Step 2**

- i. Earliest starting time of activity (i, j) is the earliest event time of the tail end event i.e. $(Es)_{ij} = E_i$
- ii. Earliest finish time of activity (i, j) is the earliest starting time + the activity time i.e. $(Ef)_{ij} = (Es)_{ij} + D_{ij}$ or $(Ef)_{ij} = E_i + D_{ij}$

- iii. Earliest event time for event j is the maximum of the earliest finish times of all activities ending in to that event i.e. $E_j = \max [(Ef)_{ij}]$ for all immediate predecessor of (i, j) or $E_j = \max [E_i + D_{ij}]$

2. Backward Pass computation (for latest allowable time)

- **Step 1**

For ending event assume $E = L$. Remember that all E's have been computed by forward pass computations.

- **Step 2**

Latest finish time for activity (i, j) is equal to the latest event time of event j i.e. $(Lf)_{ij} = L_j$

- **Step 3**

Latest starting time of activity (i, j) = the latest completion time of (i, j) – the activity time or $(Ls)_{ij} = (Lf)_{ij} - D_{ij}$ or $(Ls)_{ij} = L_j - D_{ij}$

- **Step 4**

Latest event time for event 'i' is the minimum of the latest start time of all activities originating from that event i.e. $L_i = \min [(Ls)_{ij}]$ for all immediate successor of (i, j) $= \min [(Lf)_{ij} - D_{ij}] = \min [L_j - D_{ij}]$

3. Determination of floats and slack times

There are three kinds of floats

- **Total float** – The amount of time by which the completion of an activity could be delayed beyond the earliest expected completion time without affecting the overall project duration time.

Mathematically

$$(Tf)_{ij} = (\text{Latest start} - \text{Earliest start}) \text{ for activity } (i - j)$$

$$(Tf)_{ij} = (Ls)_{ij} - (Es)_{ij} \text{ or } (Tf)_{ij} = (L_j - D_{ij}) - E_i$$

- **Free float** – The time by which the completion of an activity can be delayed beyond the earliest finish time without affecting the earliest start of a subsequent activity.

Mathematically

$$(Ff)_{ij} = (\text{Earliest time for event } j - \text{Earliest time for event } i) - \text{Activity time for } (i, j)$$

$$(Ff)_{ij} = (E_j - E_i) - D_{ij}$$

- **Independent float** – The amount of time by which the start of an activity can be delayed without effecting the earliest start time of any immediately following activities, assuming that the preceding activity has finished at its latest finish time.

Mathematically

$$(If)_{ij} = (E_j - L_i) - D_{ij}$$

The negative independent float is always taken as zero.

- **Event slack** - It is defined as the difference between the latest event and earliest event times.

Mathematically

$$\text{Head event slack} = L_j - E_j, \text{ Tail event slack} = L_i - E_i$$

4. Determination of critical path

- **Critical event** – The events with zero slack times are called critical events. In other words the event i is said to be critical if $E_i = L_i$
- **Critical activity** – The activities with zero total float are known as critical activities. In other words an activity is said to be critical if a delay in its start will cause a further delay in the completion date of the entire project.
- **Critical path** – The sequence of critical activities in a network is called critical path. The critical path is the longest path in the network from the starting event to ending event and defines the minimum time required to complete the project.

Exercise

1. What is PERT and CPM?
2. What are the advantages of using PERT/CPM?
3. Mention the applications of PERT/CPM
4. Explain the following terms
 - a. Earliest time
 - b. Latest time
 - c. Total activity slack
 - d. Event slack
 - e. Critical path
5. Explain the CPM in network analysis.
6. What are the rules for drawing network diagram? Also mention the common errors that occur in drawing networks.
7. What is the difference between PERT and CPM/

8. What are the uses of PERT and CPM?
9. Explain the basic steps in PERT/CPM techniques.
10. Write the framework of PERT/CPM.

3.1 Worked Examples on CPM

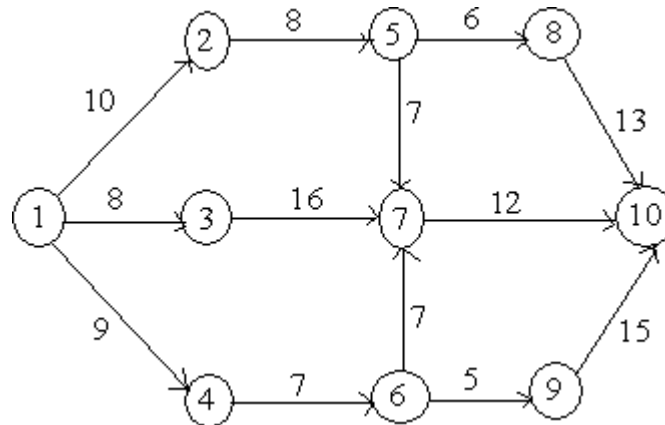
3.2 PERT

3.3 Worked Examples

3.1 Worked Examples on CPM

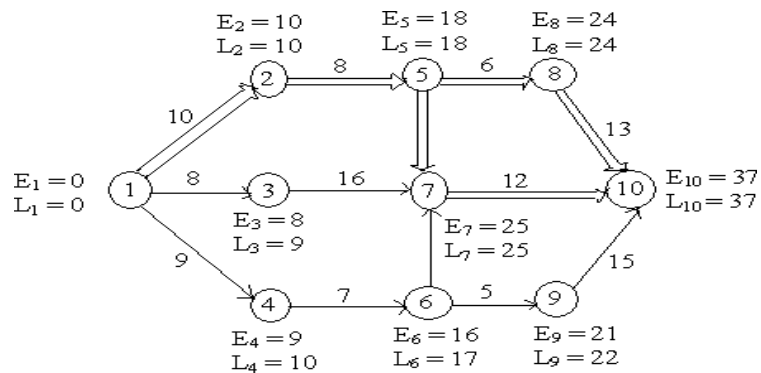
Example 1

Determine the early start and late start in respect of all node points and identify critical path for the following network.



Solution

Calculation of E and L for each node is shown in the network



Activity(i, j)	Normal Time (D _{ij})	Earliest Time		Latest Time		Float Time (L _i - D _{ij}) - E _i
		Start (E _i)	Finish (E _i + D _{ij})	Start (L _i - D _{ij})	Finish (L _i)	
(1, 2)	10	0	10	0	10	0
(1, 3)	8	0	8	1	9	1
(1, 4)	9	0	9	1	10	1
(2, 5)	8	10	18	10	18	0
(4, 6)	7	9	16	10	17	1

(3, 7)	16	8	24	9	25	1
(5, 7)	7	18	25	18	25	0
(6, 7)	7	16	23	18	25	2
(5, 8)	6	18	24	18	24	0
(6, 9)	5	16	21	17	22	1
(7, 10)	12	25	37	25	37	0
(8, 10)	13	24	37	24	37	0
(9, 10)	15	21	36	22	37	1

Network Analysis Table

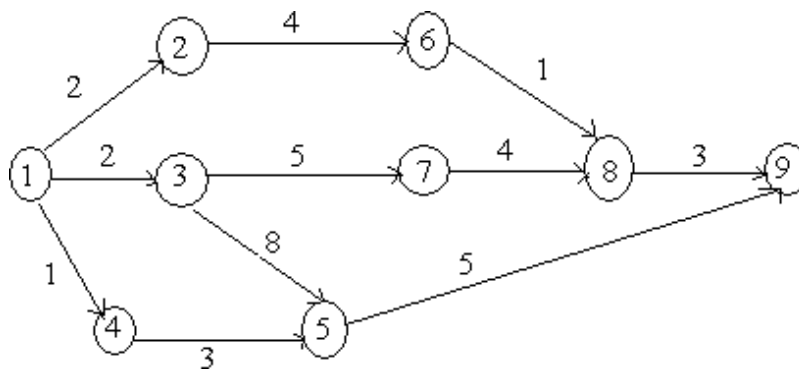
From the table, the critical nodes are (1, 2), (2, 5), (5, 7), (5, 8), (7, 10) and (8, 10)

From the table, there are two possible critical paths

- i. $1 \rightarrow 2 \rightarrow 5 \rightarrow 8 \rightarrow 10$
- ii. $1 \rightarrow 2 \rightarrow 5 \rightarrow 7 \rightarrow 10$

Example 2

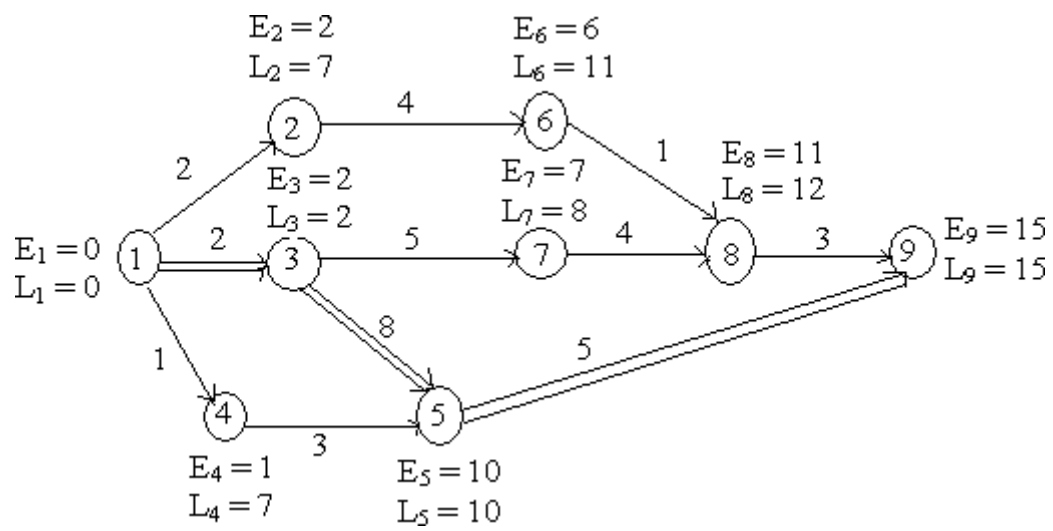
Find the critical path and calculate the slack time for the following network

**Solution**

The earliest time and the latest time are obtained below

Activity(i, j)	Normal Time (D_{ij})	Earliest Time		Latest Time		Float Time ($L_i - D_{ij}$) - E_i
		Start (E_i)	Finish ($E_i + D_{ij}$)	Start ($L_i - D_{ij}$)	Finish (L_i)	
(1, 2)	2	0	2	5	7	5
(1, 3)	2	0	2	0	2	0
(1, 4)	1	0	1	6	7	6
(2, 6)	4	2	6	7	11	5
(3, 7)	5	2	7	3	8	1
(3, 5)	8	2	10	2	10	0
(4, 5)	3	1	4	7	10	6
(5, 9)	5	10	15	10	15	0
(6, 8)	1	6	7	11	12	5
(7, 8)	4	7	11	8	12	1
(8, 9)	3	11	14	12	15	1

From the above table, the critical nodes are the activities (1, 3), (3, 5) and (5, 9)



The critical path is $1 \rightarrow 3 \rightarrow 5 \rightarrow 9$

Example 3

A project has the following times schedule

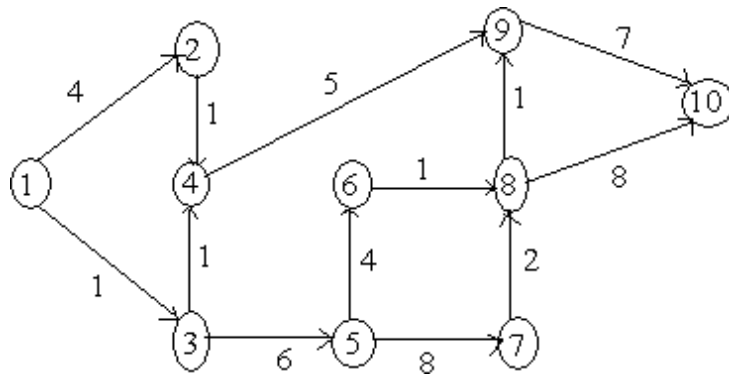
Activity	Times in weeks	Activity	Times in weeks
(1 – 2)	4	(5 – 7)	8
(1 – 3)	1	(6 – 8)	1
(2 – 4)	1	(7 – 8)	2
(3 – 4)	1	(8 – 9)	1
(3 – 5)	6	(8 – 10)	8
(4 – 9)	5	(9 – 10)	7
(5 – 6)	4		

Construct the network and compute

1. T_E and T_L for each event
2. Float for each activity
3. Critical path and its duration

Solution

The network is

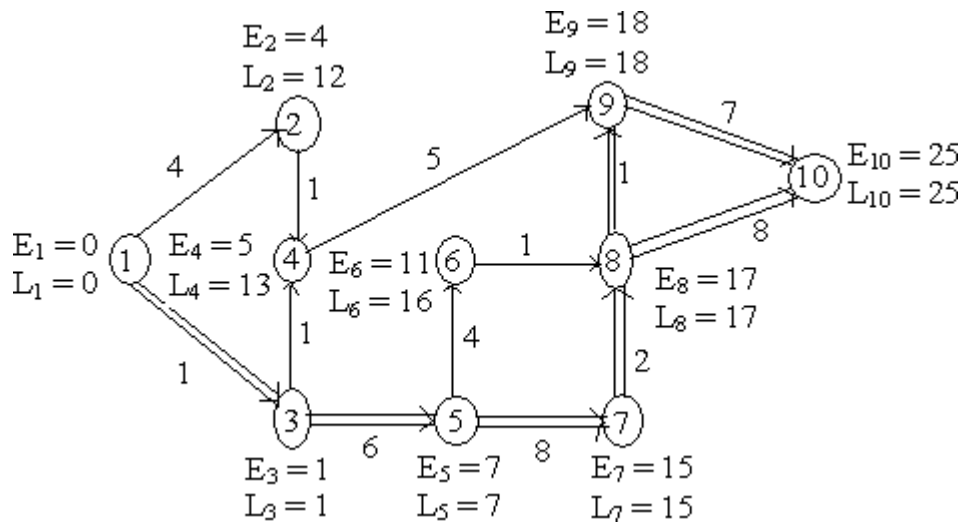


Event No.:	1	2	3	4	5	6	7	8	9	10
T_E :	0	4	1	5	7	11	15	17	18	25
T_L :	0	12	1	13	7	16	15	17	18	25

Float = T_L (Head event) – T_E (Tail event) – Duration

Activity	Duration	T_E (Tail event)	T_L (Head event)	Float
(1 – 2)	4	0	12	8
(1 – 3)	1	0	1	0
(2 – 4)	1	4	13	8
(3 – 4)	1	1	13	11
(3 – 5)	6	1	7	0
(4 – 9)	5	5	18	8
(5 – 6)	4	7	16	5
(5 – 7)	8	7	15	0
(6 – 8)	1	11	17	5
(7 – 8)	2	15	17	0
(8 – 9)	1	17	18	0
(8 – 10)	8	17	25	0
(9 – 10)	7	18	25	0

The resultant network shows the critical path



The two critical paths are

- i. $1 \rightarrow 3 \rightarrow 5 \rightarrow 7 \rightarrow 8 \rightarrow 9 \rightarrow 10$
- ii. $1 \rightarrow 3 \rightarrow 5 \rightarrow 7 \rightarrow 8 \rightarrow 10$

3.2 Project Evaluation and Review Technique (PERT)

The main objective in the analysis through PERT is to find out the completion for a particular event within specified date. The PERT approach takes into account the uncertainties. The three time values are associated with each activity

1. **Optimistic time** – It is the shortest possible time in which the activity can be finished. It assumes that every thing goes very well. This is denoted by t_0 .
2. **Most likely time** – It is the estimate of the normal time the activity would take. This assumes normal delays. If a graph is plotted in the time of completion and the frequency of completion in that time period, then most likely time will represent the highest frequency of occurrence. This is denoted by t_m .
3. **Pessimistic time** – It represents the longest time the activity could take if everything goes wrong. As in optimistic estimate, this value may be such that

only one in hundred or one in twenty will take time longer than this value. This is denoted by t_p .

In PERT calculation, all values are used to obtain the percent expected value.

1. **Expected time** – It is the average time an activity will take if it were to be repeated on large number of times and is based on the assumption that the activitytime follows Beta distribution, this is given by

$$t_e = (t_0 + 4 t_m + t_p) / 6$$

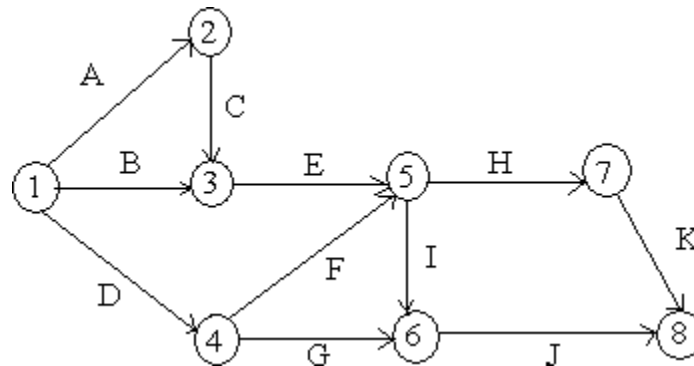
2. The **variance** for the activity is given by

$$\sigma^2 = [(t_p - t_0) / 6]^2$$

3.3 Worked Examples

Example 1

For the project



Task:	A	B	C	D	E	F	G	H	I	J	K
Least time:	4	5	8	2	4	6	8	5	3	5	6

Greatest time: 8 10 12 7 10 15 16 9 7 11 13

Most likely time: 5 7 11 3 7 9 12 6 5 8 9

Find the earliest and latest expected time to each event and also critical path in the network.

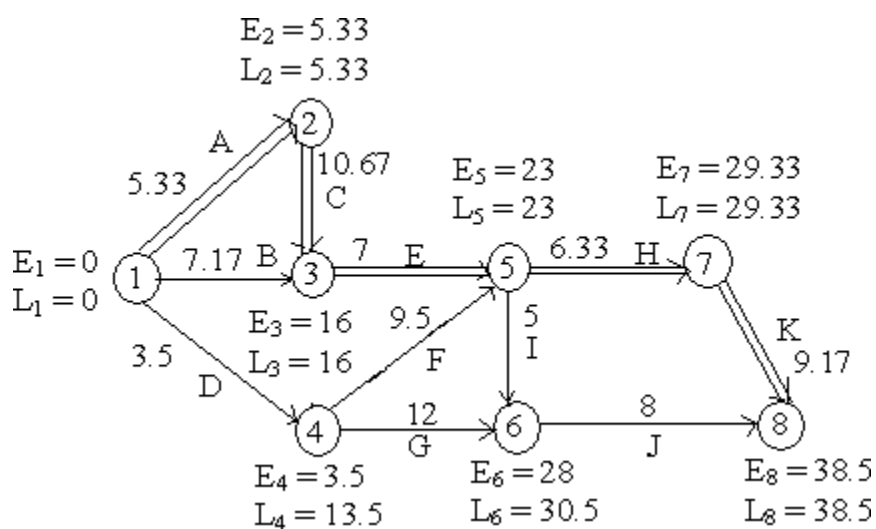
Solution

Task	Least time(t_o)	Greatest time (t_p)	Most likely time (t_m)	Expected time $(t_o + t_p + 4t_m)/6$
A	4	8	5	5.33
B	5	10	7	7.17
C	8	12	11	10.67
D	2	7	3	3.5
E	4	10	7	7
F	6	15	9	9.5
G	8	16	12	12
H	5	9	6	6.33
I	3	7	5	5
J	5	11	8	8
K	6	13	9	9.17

Task	Expected time (t_e)	Start		Finish		Total float
		Earliest	Latest	Earliest	Latest	
A	5.33	0	0	5.33	5.33	0
B	7.17	0	8.83	7.17	16	8.83
C	10.67	5.33	5.33	16	16	0
D	3.5	0	10	3.5	13.5	10
E	7	16	16	23	23	0

F	9.5	3.5	13.5	13	23	10
G	12	3.5	18.5	15.5	30.5	15
H	6.33	23	23	29.33	29.33	0
I	5	23	25.5	28	30.5	2.5
J	8	28	30.5	36	38.5	2.5
K	9.17	29.33	29.33	31.5	38.5	0

The network is



The critical path is $A \rightarrow C \rightarrow E \rightarrow H \rightarrow K$

Example 2

A project has the following characteristics

Activity	Most optimistic time (a)	Most pessimistic time (b)	Most likely time (m)
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(1 – 2)	1	5	1.5
(2 – 3)	1	3	2
(2 – 4)	1	5	3

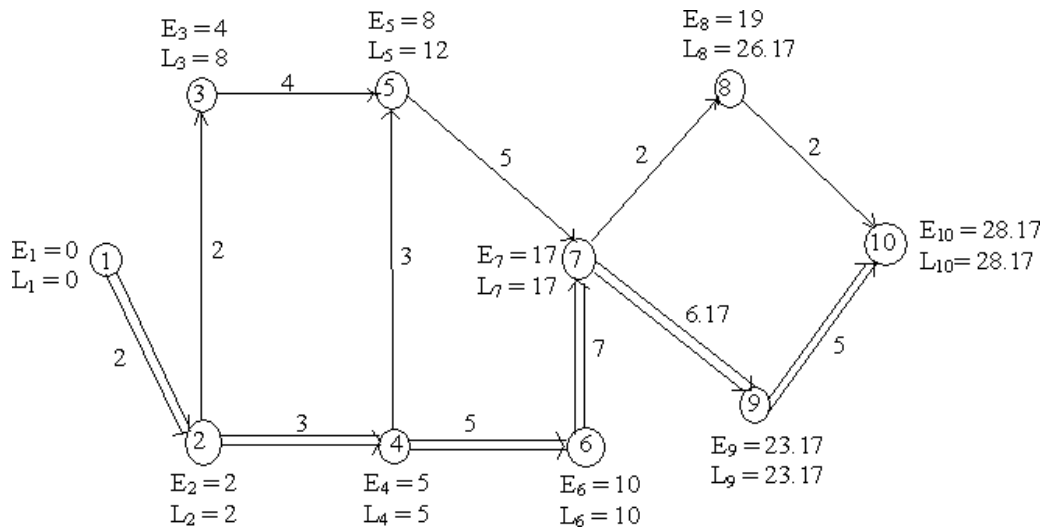
(3 – 5)	3	5	4
(4 – 5)	2	4	3
(4 – 6)	3	7	5
(5 – 7)	4	6	5
(6 – 7)	6	8	7
(7 – 8)	2	6	4
(7 – 9)	5	8	6
(8 – 10)	1	3	2
(9 – 10)	3	7	5

Construct a PERT network. Find the critical path and variance for each event.

Solution

Activity	(a)	(b)	(m)	(4m)	t_e $(a + b + 4m)/6$	v $[(b - a) / 6]^2$
(1 – 2)	1	5	1.5	6	2	4/9
(2 – 3)	1	3	2	8	2	1/9
(2 – 4)	1	5	3	12	3	4/9
(3 – 5)	3	5	4	16	4	1/9
(4 – 5)	2	4	3	12	3	1/9
(4 – 6)	3	7	5	20	5	4/9
(5 – 7)	4	6	5	20	5	1/9
(6 – 7)	6	8	7	28	7	1/9
(7 – 8)	2	6	4	16	4	4/9
(7 – 9)	5	8	6	24	6.17	1/4
(8 – 10)	1	3	2	8	2	1/9
(9 – 10)	3	7	5	20	5	4/9

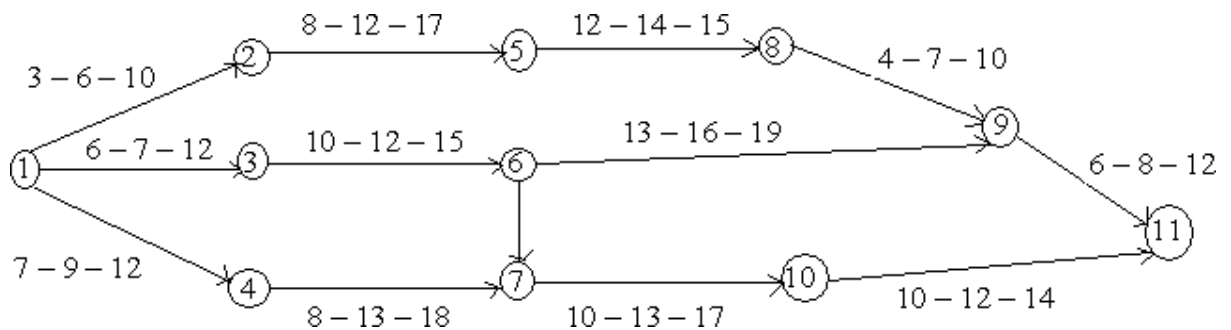
The network is constructed as shown below



The critical path = $1 \rightarrow 2 \rightarrow 4 \rightarrow 6 \rightarrow 7 \rightarrow 9 \rightarrow 10$

Example 3

Calculate the variance and the expected time for each activity



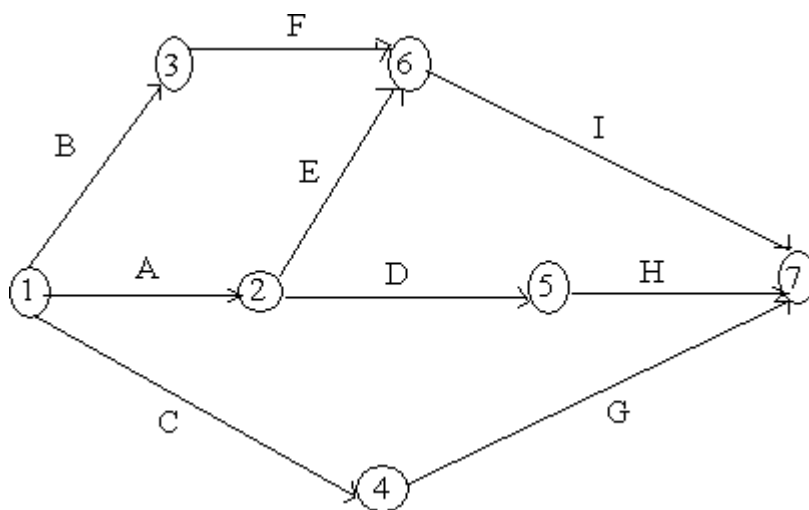
Solution

Activity	(t_o)	(t_m)	(t_p)	t_e $(t_o + t_p + 4t_m)/6$	v $[(t_p - t_o) / 6]^2$
(1 - 2)	3	6	10	6.2	1.36
(1 - 3)	6	7	12	7.7	1.00
(1 - 4)	7	9	12	9.2	0.69

(2 – 3)	0	0	0	0.0	0.00
(2 – 5)	8	12	17	12.2	2.25
(3 – 6)	10	12	15	12.2	0.69
(4 – 7)	8	13	19	13.2	3.36
(5 – 8)	12	14	15	13.9	0.25
(6 – 7)	8	9	10	9.0	0.11
(6 – 9)	13	16	19	16.0	1.00
(8 – 9)	4	7	10	7.0	1.00
(7 – 10)	10	13	17	13.2	1.36
(9 – 11)	6	8	12	8.4	1.00
(10 – 11)	10	12	14	12.0	0.66

Example 4

A project is represented by the network as shown below and has the following data



Task:	A	B	C	D	E	F	G	H	I
Least time:	5	18	26	16	15	6	7	7	3

Greatest time: 10 22 40 20 25 12 12 9 5

Most likely time: 15 20 33 18 20 9 10 8 4

Determine the following

1. Expected task time and their variance
2. Earliest and latest time

Solution

1.

Activity	Least time (t_0)	Greatest time (t_p)	Most likely time (t_m)	Expected time ($t_0 + t_p + 4t_m$)/6	Variance (σ^2)
(1-2)	5	10	8	7.8	0.69
(1-3)	18	22	20	20.0	0.44
(1-4)	26	40	33	33.0	5.43
(2-5)	16	20	18	18.0	0.44
(2-6)	15	25	20	20.0	2.78
(3-6)	6	12	9	9.0	1.00
(4-7)	7	12	10	9.8	0.69
(5-7)	7	9	8	8.0	0.11
(6-7)	3	5	4	4.0	0.11

2.

Earliest time

$$E_1 = 0$$

$$E_2 = 0 + 7.8 = 7.8$$

$$E_3 = 0 + 20 = 20$$

$$E_4 = 0 + 33 = 33$$

$$E_5 = 7.8 + 18 = 25.8$$

$$E_6 = \max [7.8 + 20, 20 + 9] = 29$$

$$E_7 = \max [33 + 9.8, 25.8 + 8, 29 + 4] = 42.8$$

Latest time

$$L_7 = 42.8$$

$$L_6 = 42.8 - 4 = 38.8$$

$$L_5 = 42.8 - 8 = 34.3$$

$$L_4 = 42.8 - 9.8 = 33$$

$$L_3 = 38.8 - 9 = 29.8$$

$$L_2 = \min [34.3 - 18, 38.8 - 20] = 16.8$$

$$L_1 = \min [16.8 - 7.8, 29.8 - 20, 33 - 33] = 0$$

1. A project schedule has the following characteristics

Activity	Most optimistic time	Most likely time	Most pessimistic time
(1 – 2)	1	2	3
(2 – 3)	1	2	3
(2 – 4)	1	3	5
(3 – 5)	3	4	5
(4 – 5)	2	5	4
(4 – 6)	3	5	7
(5 – 7)	4	5	6
(6 – 7)	6	7	8
(7 – 8)	2	4	6

